ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** cloud-free and calm	Smooth flying
2	**(c)** nitrogen	78 per cent
3	**(d)** Tamil Nadu coast	Moisture from Bay of Bengal
4	**(a)** greenhouse gases	Warming
5	**(b)** Northern Hemisphere	North lights
6	**(b)** Southern Hemisphere	South lights
7	**(a)** tropical monsoon	Monsoon type
8	**(c)** mesosphere	Friction heating
9	**(b)** thermosphere	Near poles
10	**(c)** insolation	Source of heat

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Blanket of air held by gravity (1); Mixture of gases (1).	1+1
12	Satluj (1); Beas or Ravi (1).	1+1
13	Winter and summer (1); Monsoon and post-monsoon (1).	1+1
14	Incoming solar energy (1); Decreases from equator to poles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Day: sea breeze to warm land (1); Night: land breeze to sea (1); Moderates coastal climate (1).	1+1+1
16	Troposphere and stratosphere (1); Mesosphere and thermosphere (1); Exosphere (1).	1+1+1
17	Under Ministry of Earth Sciences (1); Builds prediction models (1); Improves forecasts (1).	1+1+1
	OR: Nitrogen 78% and oxygen 21% (1); Argon 0.93% and CO2 0.04% (1); Water vapour and dust (1).	1+1+1
18	Air presses from all sides (1); Pressure inside body is equal (1); They cancel out (1).	1+1+1
	OR: Up to about 50 km (1); Calm and cloud-free (1); Contains ozone layer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Nitrogen 78%, oxygen 21% (1); Argon 0.93% and CO2 0.04% (1); Water vapour and dust (1); Shields from UV (1); Regulates temperature (1).	1+1+1+1+1
	OR: Time: day versus night (1); Cause: land warmer versus sea warmer (1); Direction: sea to land versus land to sea (1); Strength: stronger versus weaker (1); Moderate coasts (1).	1+1+1+1+1
20	Krishiparashara predicted rain using Sun and Moon (1); Brihatsamhita used nakshatras (1); Kalidasa's Meghadutam described monsoon clouds (1); Six ritus (1); Arthashastra rain measurement (1).	1+1+1+1+1
	OR: Troposphere: weather and clouds (1); Stratosphere: ozone and flights (1); Mesosphere: meteors burn (1); Thermosphere: radio waves, aurora (1); Exosphere: light gases escape (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** exosphere	Weak gravity
2	**(a)** mausim	Season
3	**(d)** carbon footprint	Definition
4	**(c)** thermosphere (ionosphere)	Charged layer
5	**(c)** 0.93%	Inert gas
6	**(d)** stratosphere	Absorbs UV
7	**(d)** low pressure	Leads to clouds
8	**(c)** climate change	Long-term
9	**(c)** tropopause	Top of troposphere
10	**(c)** from	Westerly blows from west

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Nitrogen 78% (1); Oxygen 21% (1).	1+1
12	Light display near the poles (1); Caused by solar wind particles (1).	1+1
13	Absorbs harmful ultraviolet rays (1); Protects life (1).	1+1
14	Seasonal reversal of wind direction (1); Word comes from Arabic mausim (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Vasanta, Grishma, Varsha (1); Sharad, Hemanta (1); Shishira (1).	1+1+1
16	Weather is day-to-day; climate is long-term average (1); Weather changes quickly (1); Climate is stable (1).	1+1+1
17	Insolation falls away from equator (1); Torrid, temperate and frigid zones (1); Temperature falls poleward (1).	1+1+1
	OR: Winter: December to early April (1); Summer: April to June (1); Monsoon June-September, post-monsoon October-December (1).	1+1+1
18	Agriculture depends on it (1); Water in rivers and wells (1); Floods or drought affect economy (1).	1+1+1
	OR: Heavy monsoon rain (1); Western disturbances and swollen rivers (1); Rain in upstream states (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pressure falls rapidly with height (1); Temperature falls in troposphere (1); Rises in stratosphere (1); Falls again in mesosphere (1); Rises in thermosphere (1).	1+1+1+1+1
	OR: Winter: December to April, cold (1); Summer: April to June, hot (1); Monsoon: June to September (1); Post-monsoon: October to December (1); Himalayan autumn and spring (1).	1+1+1+1+1
20	Temperature and insolation (1); Humidity (1); Precipitation (1); Atmospheric pressure (1); Wind (1).	1+1+1+1+1
	OR: Heavy monsoon rain and western disturbances (1); Satluj, Beas, Ravi swelled (1); Weak embankments and building near rivers (1); Crops and houses damaged (1); Better flood management needed (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** day-to-day state of the atmosphere	Short-term
2	**(a)** from land to sea at night	Land cools faster
3	**(a)** from sea to land in the day	Land is warmer
4	**(b)** 30 years or more	Long-term
5	**(c)** gravity	Force of attraction
6	**(c)** Satluj, Beas and Ravi	Case study
7	**(c)** June to September	Rainy season
8	**(b)** 21%	Second largest
9	**(a)** Indian Meteorological Department	Weather agency
10	**(d)** 12 km	Lowest layer

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Temperature and humidity (1); Precipitation, pressure and wind (1).	1+1
12	Total greenhouse gas emitted by activities (1); Of a person or product (1).	1+1
13	Weather is short-term (1); Climate is the average over 30 years or more (1).	1+1
14	Contains air we breathe and water vapour (1); All weather happens here (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	SW: June-Sept, sea to land, most rain (1); NE: Oct-Feb, land to sea, dry (1); Rain to east coast (1).	1+1+1
16	Burning fossil fuels (1); Deforestation (1); Industrial pollution (1).	1+1+1
17	Weak embankments (1); Building near rivers (1); Silt and late warnings (1).	1+1+1
	OR: Warm air holds more vapour (1); Sweat evaporates slowly (1); We feel uncomfortable (1).	1+1+1
18	Warm air rises giving low pressure and clouds (1); Cold air sinks giving high pressure (1); Clear and sunny weather (1).	1+1+1
	OR: Save energy (1); Use public transport (1); Plant trees (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Light display near poles (1); Solar wind particles guided by magnetic field (1); Gases glow (1); Borealis and Australis (1); In the thermosphere (1).	1+1+1+1+1
	OR: Trap heat in the atmosphere (1); CO2, methane, nitrous oxide (1); Raised by human activity (1); Global temperatures rise (1); Climate shifts (1).	1+1+1+1+1
20	Seasonal reversal of winds (1); Land heats faster giving low pressure (1); SW monsoon brings most rain (1); NE monsoon brings rain to east coast (1); Importance for agriculture and economy (1).	1+1+1+1+1
	OR: Caused by fossil fuels and deforestation (1); Greenhouse gases rise (1); Floods, droughts, melting glaciers (1); Save energy, use renewables (1); Plant trees (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.